Principled Convergence, Operational Fragmentation: Healthcare AI Governance in the US, EU, China, and Africa

Kingsley Mbewe[1,2], Por Srenhong[1], Zaineb El Hamdou[1], Wighio Kainat[1,3], Zheng Jiang, PhD[1*], Shaffi Mdala, MD[4,5], Yi Wang, PhD[1]

[1] Beijing University of Posts and Telecommunications, Beijing, China
[2] University of Malawi, Zomba, Malawi
[3] University of Sindh, Jamshoro, Pakistan
[4] Queen Elizabeth Central Hospital, Blantyre, Malawi
[5] Kamuzu University of Health Sciences, Blantyre, Malawi

***Corresponding Author:**
Zheng Jiang
Beijing University of Posts and Telecommunications
Email: jiangzheng@bupt.edu.cn
Telephone: +86-10-6231-7908

Summary

The integration of artificial intelligence (AI) into global healthcare workflows is advancing faster than state governance systems can adapt around the world. This Health Policy paper shifts beyond the traditional Global North versus Global South binary, providing a comparative regulatory analysis of a landscape divided between dominant technology-exporting superpowers (the United States, the European Union, and China) and technology-importing regions (the African continent). While exporting powers demonstrate principled convergence on safety, lifecycle management, and human oversight, their underlying legal architectures introduce severe operational fragmentation. For technology-importing regions, this fragmentation creates systemic vulnerabilities, exposing local populations to regulatory arbitrage and asymmetric dependencies on foreign cloud infrastructure. This operational friction has triggered an emergent narrative of Sovereign AI, wherein developing nations prioritize data sovereignty and localized compute infrastructure to guarantee algorithmic alignment with regional clinical realities. Ultimately, mitigating global health disparities requires reconciling fragmented state legal regimes with the normative ethical frameworks championed by intergovernmental and non-governmental institutions, shifting toward a unified, human-centered global regulatory science.

Search strategy and selection criteria:

Regulatory frameworks, binding legislation, and normative policy instruments were identified through systematic tracking and searches of official governmental portals, legal databases, and institutional repositories up to June, 2026. Primary source repositories included the US Food and Drug Administration (FDA) regulatory database, the European Union law repository (EUR-Lex), the National Medical Products Administration (NMPA) official digital archive, the African Union portal, and the World Health Organization (WHO) Institutional Repository for Information Sharing (IRIS).

To ensure exhaustive data capture, structured Boolean search strings were deployed across EUR-Lex and the WHO IRIS in April 2026. Search terms included "artificial intelligence", "machine learning", "medical device", "healthcare regulation", "data privacy", "health equity", and "ethics." All returned results underwent a rigorous human check to verify substantive relevance and minimize selection bias. To systematically map the African regulatory landscape, the official governmental files and digital ministry portals of all 54 African countries were comprehensively audited, with the policy documents curated by AI Policy Lab (`https://www.aipolicy.africa/`) as the complementary source. This manual human check revealed that only 22 countries have published relevant governmental files, AI policies, or national strategies; the remaining 32 countries yielded no official documentation.

Additional ethical and normative frameworks were retrieved directly from the Holy See digital archives and international medical association databases. Only official legislative acts, finalized agency guidance documents, technical industry standards, and authoritative institutional declarations relevant to healthcare were selected for analysis. Draft proposals subsequently superseded by finalized frameworks, or non-sector-specific general AI strategies, were excluded.

Key message:

1. *A New Geopolitical Paradigm*: The global healthcare AI landscape defies the traditional Global North versus Global South binary; it is instead structurally divided into dominant technology-exporting superpowers (the US, EU, and China) and technology-importing regions.

2. *The Paradox of Operational Fragmentation*: Despite a principled convergence on core ethics like safety and human oversight, severe operational fragmentation across US, EU, and Chinese legal architectures inhibits seamless cross-border clinical validation and creates profound compliance friction.

3. *Vulnerability and Sovereign AI*: The absence of operative continental approval pathways leaves the African continent highly vulnerable to regulatory arbitrage. In response, technology-importing nations are adopting "Sovereign AI" strategies that position localized digital infrastructure and data sovereignty as fundamental to national health security.

4. *The Power of Non-State Actors*: Non-governmental normative frameworks, such as the World Health Organization's technical guidance and faith-based institutional ethics like the Magnifica Humanitas encyclical, provide vital moral infrastructure that establishes baseline accountability where formal state regulations fail.

5. *A Collaborative Policy Path*: Mitigating global health disparities requires moving beyond isolated domestic mandates through multilateral collaborations, including extraterritorial accountability for tech-exporters, unified continental review mechanisms for importers, and cross-border regulatory sandboxes.

Introduction

The integration of artificial intelligence (AI), particularly the recent progress of generative AI, into clinical infrastructure is transitioning from experimental spaces to operational reality at a velocity that severely outstrips the evolution of governance frameworks.[1,2] At the same time, this regulatory landscape defies a simplistic monolithic structure or a traditional Global North versus Global South binary.[3,4] Instead, the global ecosystem is characterized by a stark division between dominant technology-exporting jurisdictions and technology-importing regions.[5] The United States, the European Union, and China have emerged as three primary regulatory superpowers, each projecting distinct legal philosophies across international borders.[6–8] While these powers share high-level ethical commitments to safety and transparency, they diverge sharply on core operational mechanisms, including data privacy architectures, liability allocation, and adaptive change management protocols. For instance, the EU's General Data Protection Regulation (GDPR) treats health data as a strictly restricted special category,[9] whereas the US Health Insurance Portability and Accountability Act (HIPAA) provides flexible exemptions that permit broad commercial data sharing for clinical operations.[10]

For technology-importing regions, particularly across the African continent, this operational fragmentation introduces profound vulnerabilities.[11,12] Lacking binding continental approval mechanisms or mature domestic risk-classification frameworks, these regions face the threat of regulatory arbitrage, where unsafe or unvalidated algorithms are deployed in thin governance environments.[13–15] Furthermore, as low-resource health systems become dependent on foundational models and cloud computing infrastructure controlled by a handful of overseas vendors, a critical loss of technological self-determination occurs. This infrastructure gap has triggered an emergent counter-narrative of Sovereign AI, framing data sovereignty and localized compute infrastructure as core pillars of national health security.[16,17] This infrastructure gap has triggered an emergent counter-narrative of Sovereign AI. Developing national strategies, such as Kenya's 2025–2030 framework, are increasingly positioning data sovereignty and localized compute infrastructure as core pillars of national health security, arguing that a nation must control its own data assets to build tools tailored to its unique clinical and linguistic realities.[18]

Therefore, this health policy paper provides a comprehensive, comparative analysis of the legal and ethical architectures governing healthcare AI across the US, the EU, China, and the African continent. By examining how formal state mechanisms intersect with influential non-governmental normative frameworks—including the World Health Organization's (WHO) technical guidance[19] and the structural critiques of the 2026 papal encyclical Magnifica Humanitas,[20] we characterize the multi-layered ecosystem of modern health governance. Ultimately, we demonstrate that while the international community has achieved a principled convergence on what constitutes ethical AI, it remains trapped in a state of severe operational fragmentation that threatens global health equity and clinical sovereignty.

Technological Acceleration and the Governance Lag

The integration of intelligent systems into clinical infrastructure is transitioning from experimental spaces to operational reality at an unprecedented pace. Health and medical AI is undergoing a profound paradigm shift from passive, isolated tools to autonomous, active agentic collaborators, with multi-agent frameworks now simulating complex multidisciplinary clinical expert discussions and scalable foundation models transforming fields like digital pathology. In response, technology-exporting jurisdictions have accelerated market clearance; the US Food and Drug Administration (FDA) has cleared over 1,200 AI-enabled devices, while China's National Medical Products Administration (NMPA) has authorized over 110 Class III deep-learning software products.[21,22]

However, the emerging artificial intelligence should be viewed as a structural means to an

end rather than an absolute panacea for the complex systemic challenges facing global healthcare.[23–25] This rapid proliferation exposes a fundamental governance lag. Traditional medical device regulations were engineered for static, hardware-adjacent products evaluated at a single point in time. Contemporary health AI architectures, by contrast, are inherently iterative, highly adaptive, and capable of post-market drift. Current state oversight systems struggle to manage this velocity. Furthermore, as highlighted by a recent editorial in The Lancet Digital Health, medical AI is undergoing a profound paradigm shift from passive, isolated tools to autonomous, active agentic collaborators.[26] Multi-agent AI frameworks are now being developed to simulate complex, multidisciplinary clinical expert discussions, such as transplant selection committees, and to autonomously manage end-to-end diagnostic workflows.[27,28] Simultaneously, the clinical deployment of highly scalable vision-language foundation models is transforming fields like digital pathology. They process whole slide images (WSIs) to identify molecular alterations and detect cancer biomarkers, radically altering the clinical landscape. However, these technological progresses are introducing severe operational, ethical, and legal challenges that current regulatory architectures are not well-equipped to handle.[29,30] The real-world deployment of multi-agent systems and foundation models is deeply hampered by data heterogeneity, systemic information silos, and fragmentation across clinical institutions.[31,32] Furthermore, as AI agents take on active decision-making roles within multidisciplinary healthcare teams, the legal boundaries of professional accountability become dangerously obscured, necessitating a complete rethink of the clinician's role from a visual analyst to an inspector providing structural human oversight.[33–35]

When governance frameworks fail to keep pace with continuous model updates and data drift, the real-world consequences manifest immediately in global practice.[36,37] Because AI models learn from historical datasets, they inevitably inherit and amplify the systemic biases embedded within those data structures.[38,39] When diagnostic algorithms or clinical risk models are trained on demographically unrepresentative populations, or rely on flawed socioeconomic proxies, such as utilizing institutional healthcare spending as a surrogate for actual medical need, they systematically misallocate resources and widen health disparities for marginalized populations.[40] While foundation models offer an extraordinary opportunity to democratize precision healthcare and improve equitable diagnostic access in limited-resource settings where expensive genomic testing is unavailable, they also require rigorous continuous bias assessments, external validation across diverse multi-institutional datasets, and robust safeguarding policies.[41–43] Without continuous lifecycle oversight, dynamic change-control protocols, and multi-institutional validation pathways, the widening lag between technological deployment and regulatory oversight directly compromises patient safety and clinical reliability.

Dominant Technology-Exporting Regulatory Architectures

The global landscape of AI governance is largely shaped by three dominant regulatory superpowers: the United States (US), the European Union (EU), and China. While these technology-exporting jurisdictions share core principles, for instance, risk-stratified classification, the necessity of representative training data, and mandatory human oversight, their legal instruments and operational mechanisms for managing adaptive AI diverge significantly.

The United States: Lifecycle Adaptability and Commercial Facilitation

The US utilizes a flexible, product-specific framework overseen by the Food and Drug Administration (FDA) under the Federal Food, Drug, and Cosmetic Act (FDCA) established in 1938. FDA follows its convention of viewing Software as a Medical Device (SaMD). Thus, rather than enacting comprehensive new AI legislation, the FDA relies on its traditional three-tier risk classification: Class I (low risk), Class II (moderate risk, requiring 510(k) clearance or De

Novo classification), and Class III (high risk, requiring Premarket Approval [PMA]) [44,45]. This commercially facilitative approach has yielded rapid market entry; as of August 2025, the FDA had cleared or approved over 1,200 AI-enabled medical devices, comprising 16 PMAs, 36 De Novo classifications, and 1,195 devices cleared through the 510(k) pathway.

To accommodate the inherently iterative nature of machine learning, the FDA centers its oversight on a Total Product Lifecycle (TPLC) approach, operationalized through the Predetermined Change Control Plan (PCCP). [46] Formalized in a December 2024 final guidance document, the PCCP allows manufacturers to pre-specify a Modification Description, establish a Modification Protocol, and conduct an Impact Assessment prior to market deployment. [44] Once the FDA approves the PCCP, manufacturers can implement specified post-market modifications without submitting new premarket applications. This mechanism is profoundly critical for the deployment of agentic AI systems and scalable foundation models, providing the regulatory flexibility required for these systems to continuously learn from new clinical data.

Furthermore, the FDA has partnered with Health Canada and the UK's Medicines and Healthcare products Regulatory Agency (MHRA) to establish Good Machine Learning Practice (GMLP) principles, which was later adopted by the International Medical Device Regulators Forum (IMDRF) in 2025. [47] However, because the US system relies heavily on agency guidance rather than statutory law, the burden of ensuring multidisciplinary expertise, model explainability, and continuous bias assessment falls largely on the manufacturer.

The European Union: Precautionary Legislation and Fundamental Rights

In contrast, the EU has constructed the world's first comprehensive, binding, cross-sectoral AI legislation. The EU AI Act (AIA 2024/1689), which entered into force in August 2024, operates under a precautionary, rights-based philosophy. [48–50] The Act classifies AI into four tiers (unacceptable, high-risk, limited, and minimal risk), automatically categorizing AI-enabled medical tools as "high-risk." This high-risk classification triggers rigorous statutory demands. Under Article 10, developers must govern training datasets to ensure demographic representativeness and mitigate systemic bias. Under Article 14, systems must be explicitly designed for effective human oversight, allowing clinicians to interpret and override outputs. This legislative demand directly supports the clinical imperative highlighted in recent Lancet Digital Health editorials: as AI takes on agentic roles in complex diagnostic workflows like pathology, the human expert must transition from a visual analyst to an active overseer to detect and override automation bias produced by algorithms.

A major operational complexity within the EU is the intersection of the AI Act with the existing Medical Device Regulation (MDR 2017/745) and In Vitro Diagnostic Medical Device Regulation (IVDR 2017/746). [51,52] To clarify this, the Joint Artificial Intelligence and Medical Device Coordination Group issued guidance (MDCG 2025-6) in September 2025. [53] Under Article 8(2) of the AI Act, manufacturers can integrate AI risk management, quality systems, and post-market monitoring into their existing MDR/IVDR documentation. Thus, the new legislative framework requires a simultaneous and complementary application of the MDR/IVDR and the AIA for medical devices that contains one or more high-risk AI system. Manufacturers of these high-risk medical AI systems face a strict compliance deadline of August 2, 2027. Furthermore, the EU's forthcoming AI Liability Directive, which presumes causality when a developer's non-compliance contributes to a patient injury, represents a major structural shift in accountability allocation, a vital safeguard as AI participate in multidisciplinary clinical and healthcare practices. [54]

China: Structured Rules and Industrial Scale

China's framework, governed by the National Medical Products Administration (NMPA), occupies a unique middle ground: it combines highly structured technical standards with aggressive

industrial policy aimed at moving approved products into clinical use quickly.[55,56] The NMPA manages Class II and Class III AI devices through rigorous registration pathways, having issued 19 specific guidance documents, including the world's first deep learning-assisted review guidelines.[57] These documents forms comprehensive rules that place a heavy emphasis on industrial standardization. An Artificial Intelligence Medical Device Standardization Technical Committee (SMD/TU 002) was established in 2019, had issued eight industry standards by June 2025. By October 2025, the NMPA had approved more than 110 deep learning-based Class III medical device software products covering radiology, ophthalmology, pathology, and cardiology. While requiring stringent algorithm transparency and detailed version tracking for post-market lifecycle management, the NMPA also offers expedited Priority Review pathways for "domestic first, world-leading" innovations with high clinical value.

Crucially, China actively pursues international regulatory alignment, referencing documents from the International Medical Device Regulators Forum (IMDRF), the Global Harmonization Working Party (GHWP), and the ITU/WHO Focus Group on AI for Health, As the NMPA declared to "fully references international regulatory experience."[58] This dual strategy of accelerated domestic innovation and international alignment not only accelerates the integration of advanced foundation models into routine clinical workflows but also strategically positions China to export its standardized AI infrastructure and products to technology-importing regions across the Global South.[59,60]

Commonalities and Divergences Across the US, the EU, and China

The three jurisdictions share a set of core commitments. These are not vague aspirations; they show up as concrete requirements in each system, even if the concrete mechanism may vary. Meanwhile, five divergences stand out, each with practical consequences for any organization operating across more than one of these markets. Panel 1 summarizes these key commonalities and divergences.

***Panel 1*: Overview of commonalities and divergences across the US, the EU, and China**

Key Commonalities

- **Risk-stratified classification.** All three tie regulatory intensity to clinical risk. The EU's four-tier AI Act, the FDA's three-class device framework, and China's Class II/III NMPA pathways reach the same practical conclusion: an AI tool that influences a cancer diagnosis gets harder scrutiny than one that schedules appointments. The threshold differs; the logic does not.
- **Lifecycle oversight beyond point-of-approval.** None of the three treats regulatory clearance as a finish line. AI systems keep evolving; a model approved today may behave differently after six months of real-world retraining. The EU's ongoing risk management requirements, the FDA's Total Product Lifecycle framework and PCCP mechanism, and China's version-tracking obligations all reflect the same recognition: you cannot approve something once and walk away.

(Panel 1 continued from previous page)

- **Training data quality and representativeness.** Each framework requires, at varying levels of detail, that training data actually represent the patients the system will be used on. The EU AI Act's Article 10 data governance rules, FDA's GMLP principles, and NMPA's algorithm transparency requirements all stem from the same problem: a model trained mostly on data from one demographic group will perform worse on everyone else. That is not a technical edge case; it is a documented failure mode with real clinical consequences.
- **Human oversight as a non-negotiable safeguard.** All three draw a hard line against AI making consequential clinical decisions without a human in the loop. The EU AI Act's Article 14, the FDA/MHRA's GMLP human-centred design principle, and NMPA's transparency mandates converge on this. A clinician must be able to understand what the system said, why, and override it. No jurisdiction has been willing to drop that requirement, even for systems with strong performance metrics.
- **Transparency and explainability as design requirements.** The three differ in how hard they push on this, but none treats it as optional. The EU makes explainability a binding requirement for high-risk systems. The US builds it into GMLP guidance and pushes developers toward interpretable outputs through performance-based standards. China requires manufacturers to document how their systems reason and to make that explanation available to both clinicians and regulators. The gap is in enforcement, not in the principle itself.
- **International regulatory collaboration.** All three jurisdictions cite IMDRF and GHWP reference documents and participate in harmonization efforts. The trilateral FDA-Health Canada-MHRA GMLP project is the clearest example of this in practice: three regulators co-producing a shared framework rather than each writing their own. These efforts are real, even if binding international standards remain out of reach for now.

Key Divergences

- **Regulatory philosophy: precaution versus facilitation.** The EU puts the burden on the developer to prove safety before deployment. The AI Act's grounding in human dignity and fundamental rights sets a ceiling on what AI systems can do regardless of how well they perform technically. The US starts from a different default: the presumption runs toward allowing market entry once safety and effectiveness thresholds are met, and specific mechanisms like the PCCP exist to reduce friction for companies that want to keep improving their products post-clearance. China stands somewhere in between, technically demanding but oriented toward moving approved products into clinical use quickly and building domestic AI industry capacity, which is never made explicit by the EU and the US.

(Panel 1 continued from previous page)

- **Regulatory instrument and legal architecture.** The US runs on agency guidance. The EU passed a statute. China works through technical standards and registration requirements. A statute gives people legal certainty about what is required and what the consequences of non-compliance are. Guidance documents are easier to revise but carry less weight and can shift without notice. For a manufacturer trying to plan a market entry strategy five years out, the US framework is familiar to industry but fluid; the EU framework is harder to navigate initially but more predictable once understood; and China's framework, being agency-driven, tends to evolve at pace with government priorities rather than on a legislative timeline.

- **Data privacy and health data status.** This is the divergence with the most immediate operational effect. Under EU's GDPR, collecting health data for AI development requires explicit consent or a clearly defined lawful basis.HIPAA is more permissive. Data can be shared for treatment, payment, and healthcare operations without individual consent, enabling companies like Google can legally build healthcare AI on hospital records at scale. The 2019 Project Nightingale episode illustrated the gap between what is permitted and what patients expect. China's Personal Information Protection Law requires consent as a general rule, but how that applies specifically to AI training pipelines is still in absence. Meanwhile, cross-border data transfers face additional restrictions that complicate multinational development. In practice, the same dataset can be legal in one jurisdiction but not in another, a genuine compliance problem with no clear solution.

- **Management of adaptive AI and post-approval change.** This is where the frameworks diverge most sharply. The FDA's PCCP is the most operationalizable solution so far. A manufacturer describes planned changes in advance, gets them pre-approved, and can then implement them without filing a new submission each time, offering a practical way to deal with AI systems that learn and drift over time. The EU AI Act requires ongoing risk management and mandates reassessment when significant changes occur, but has not yet created a clear pre-authorisation path. China draws a line between major and non-major changes, with the latter handled through the manufacturer's own quality system rather than formal re-registration.The FDA's approach is the most developer-friendly; the EU's offers more predictable legal boundaries; China's is the least burdensome for straightforward updates. None of them fully resolves the underlying tension between regulatory certainty and AI's tendency to keep changing.

- **Liability and accountability allocation.** No jurisdiction has sorted this out yet, but the starting positions are very different. The EU is working on an AI Liability Directive that would make it easier for harmed patients to hold developers accountable: where a developer's non-compliance contributed to an injury, causation would be presumed rather than requiring the patient to prove it. That is a meaningful shift in who bears the burden of proof. The US leaves these questions to existing tort law and medical malpractice doctrine, where the clinician who acted on an AI recommendation usually ends up holding the liability – a familiar allocation for hospitals and practitioners, but one that does not obviously reach AI developers. China has not yet developed AI-specific liability rules, leaving both domestic and foreign manufacturers with genuine uncertainty about their exposure. When the same AI system causes the same type of error in all three markets, the legal consequences are likely to be entirely different.

Technology-Importing Regions and Sovereign AI Narratives

While the operational fragmentation of global AI governance impacts technology-importing regions across the entire Global South, this analysis deliberately centers on the African continent, since the gap between articulated ethical principles and enforceable regulatory capacity is wider in Africa than in any other region examined.[61,62] Unlike parts of Latin America or Southeast Asia, which increasingly benefit from established regional health cooperation pathways, maturing domestic regulatory agencies, and localized digital infrastructures, Africa faces a unique convergence of infrastructural and governance deficits.[63,64] Currently, the continent relies almost entirely on external cloud infrastructure and foreign foundational models, creating a profound structural vulnerability, while their citizens are locked in the lower-end of AI value chain by performing ghost AI jobs such as data annotations.[65] Furthermore, a 2025 Nature Health analysis concluded that no single national medical device regulator on the African continent has yet reached the maturity level required to independently assess AI-enabled medical devices.[66] This combination of fragmented national strategies, the lack of an operative continental approval mechanism, and an acute infrastructural dependency makes Africa the ideal case study for the risks of regulatory arbitrage and the urgency of Sovereign AI.

Table 1: Comparative Landscape of National AI Strategies and Clinical Focus Areas in Africa

Country	**Status**	**Year**	**Healthcare**	**Focused Areas**	**Notable Initiative**
Benin[67]	Full strategy	2023	Yes*	Supporting ethical, inclusive & sovereign AI	International collaboration of AGI.HEALTH
Ghana[68]	Full strategy	2023	Yes*	Chest condition diagnostics, Deep Science health AI, Bias mitigation	minoHealth AI Labs – diagnoses 14 chest conditions
Rwanda[69]	Full strategy	2023	Yes*	Diagnostics, Health info systems, Clinician training, Ethical oversight	"AI for Health" strategy; ethical oversight committees
Ethiopia[70]	Full strategy	2024	Yes*	Pediatric care, Digital health literacy, General-purpose AI	iCog Labs; AU High-Level AI Policy Dialogue host
Nigeria[71]	Full strategy	2025	Yes*	Public-private partnerships, Data governance, NLP for clinical settings	Intron Health – NLP for African accents in clinics
Egypt[72,73]	Full strategy	2025	Yes*	SDG-aligned health goals, Feasibility-based AI use case selection	AI Factory with Nvidia & Cassava Technologies
Zimbabwe[74]	Full strategy	2026	Yes*	AI for inclusive development, shared prosperity, and citizen dignity	Healthcare as one of three national priority areas
Mauritius[75]	Full strategy	2018	Partial**	Digital health, eHealth strategy	National eHealth Strategy; policy continuity
Tunisia[76]	Full strategy	2019	Partial**	Public sector health reform, AI talent pipeline	Highest developer density in Africa (4,120/million)
Senegal[77]	Full strategy	2023	Partial**	Digital health, AI laws & regulation	Satellite internet & 5G for AI access
Morocco[78]	Full strategy	2024	Partial**	Health sector AI implementation, Public sector reform	National AI centres approved by government

Zambia[79]	Full strategy	2024	Partial**	Health in SDG targets, Priority sector rollout	Partnership between (CIDRZ) and Qure.ai for TB screening
Cameroon[80]	Full strategy	2025	Partial**	Public sector AI reform	12 sovereign, high-impact AI solutions
Côte d'Ivoire[81]	Full strategy	2024	Partial**	Newly entered AI policy space, Health as development goal	Creating National Committee for AI and Data Governance
Kenya[18]	Draft/in progress	2025	Yes*	Healthcare priority sector, Data sovereignty, Health diagnostics	Maternal-health chatbot & disease advisory system; Konza Technopolis AI hub
South Africa[82,83]	Draft/in progress	2024	Partial**	Public sector health reform, African language NLP	Lelapa AI – NLP for African languages
Tanzania[84]	Draft/in progress	Draft	Partial**	Public sector AI reform, Health services	Draft strategy; eHealth systems underway
Uganda[85]	Draft/in progress	Draft	Partial**	Public sector AI reform, Health services	AI for Universal Health Coverage
Botswana[86]	Draft/in progress	Draft	N/A***	Ethical, inclusive and secure AI adoption	–
Namibia[87]	Draft/in progress	Draft	No	key public sectors like agriculture, healthcare, and education	–

*. Yes — explicitly named: The strategy or policy explicitly identifies health or healthcare — for example, naming it as a priority sector or setting out dedicated health use cases, objectives, or governance measures.
**. Partial — implicitly included: Health is not singled out as a distinct focus but is covered indirectly — for instance through general references to priority sectors, public-service delivery, or SDG-aligned goals — without health-specific commitments.
***. N/A: The strategy or policy contains no explicit or implicit reference to AI applications in healthcare.

The Governance Vacuum and the Threat of Regulatory Arbitrage

Across the continent, formal AI governance remains a patchwork of nationally diverse, largely non-binding advisory strategies rather than enforceable statutory law. At the supranational level, the African Union Executive Council adopted the Continental Artificial Intelligence Strategy in July 2024; however, this framework is non-binding, relies on phased implementation through 2030, and establishes no specific obligations for healthcare.[88] Similarly, while the African Medicines Agency (AMA) treaty entered into force in November 2021 and 32 nations have ratified the treaty, the body has yet to establish an operational, cross-border continental mechanism for AI medical device approval.[89]

Below this continental level, most countries have strong interest in developing AI policies, evidenced by the broad participation of UNESCO's AI Readiness Assessment Methodology (RAM)[90]. However, only 20 African countries have adopted or are actively drafting national AI policies (see Table 1) by May 2026, demonstrating varying degrees of institutional commitment. For example, Rwanda established the continent's first dedicated National AI Agency in June 2025, and Ghana recently updated its National AI Strategy for 2025–2035 with healthcare explicitly designated as a priority sector. Despite these vital political commitments, no African nation has successfully implemented binding risk-stratified classification architectures, predetermined change control mechanisms, or formal post-market surveillance protocols for adaptive

algorithms specifically designed for healthcare. Even the policy-makers themselves often use AI improperly. For instance, South Africa's Draft South Africa National Artificial Intelligence Policy was withdrawn for citing a number of fictitious research publications created by AI's hallucination, offering ironic evidence of regulation vacuum.[83] This vacuum exposes the continent to severe regulatory arbitrage, creating an environment where black-box algorithms that fail rigorous demographic validation in dominant exporting jurisdictions might be legally deployed in African markets with minimal oversight. Moreover, scheduled legalization processes in multiple countries are often delayed, for example, Uganda has not yet published its national AI governance framework which was scheduled in 2025.

Infrastructure Dependency and the Sovereign AI Counter-Narrative

In direct response to this regulatory asymmetry, a powerful counter-narrative of Sovereign AI is materializing, driven by the realization that global health equity requires both data sovereignty and localized computing infrastructure. African policymakers are increasingly recognizing that utilizing diagnostic algorithms hosted on multinational servers exposes domestic clinical systems to foreign data extraction and arbitrary disruptions in service terms.[91] Consequently, emerging legislative frameworks, such as Kenya's Artificial Intelligence Strategy 2025–2030 (launched in March 2025), place an explicit policy emphasis on data sovereignty and domestic infrastructure control.[18]

To physically anchor this sovereignty, large-scale commercial initiatives are beginning to deploy localized computing power, most notably the $720 million Cassava-Nvidia AI Factory, which launched its first large-scale GPU-as-a-service facility in South Africa in March 2026, with planned expansions into Nigeria, Kenya, Egypt, and Morocco.[92] Empowered by this shifting infrastructure, domestic innovators are building and validating foundation models trained specifically on African clinical realities. Nigeria's Intron Health has compiled a massive clinical speech database encompassing over 3.5 million audio clips across 200 African accents to ensure voice-integrated electronic medical records function for local populations,[93] while South Africa's Lelapa AI is actively developing natural language processing models (such as InkubaLM) for low-resource languages like Swahili, Yoruba, and isiZulu.[94] Similarly, Ghana's minoHealth AI Labs has achieved remarkable performances in diagnosing 14 chest conditions via deep learning in local cohorts, outperforming individual human radiologists.[95]

These advancements demonstrate that for technology-importing regions, mitigating the risks of imported AI is not merely a bureaucratic exercise in writing stricter regulations; it is a fundamental assertion of technological self-determination, ensuring that digital medicine serves, rather than extracts from, vulnerable populations.

The Role of Non-Governmental Normative Frameworks

While formal state regulations are predominantly preoccupied with technical conformity and rapid market clearance, they frequently neglect the foundational ethical question of what kind of medicine AI ought to enable. In the absence of a unified global regulatory authority, non-governmental actors such as intergovernmental agencies, international medical associations, and global faith-based health networks, serve as a vital moral and operational infrastructure. For example, the World Medical Association's Declaration of Helsinki is actively interpreted by national medical bodies to require informed consent for patient data used in AI training,[96,97] while professional societies globally mandate that AI remain a clinical decision-support tool rather than an autonomous actor.[98,99] These entities establish baseline imperatives for equity and accountability that often precede and inform binding state legislation.

The World Health Organization: Standardizing Global Norms

The World Health Organization (WHO) operates as the most consequential intergovernmental actor in this space, leveraging its normative authority to translate ethical consensus into actionable policy for Member States. Central to this effort is the 2021 Ethics and Governance of Artificial Intelligence for Health guidance, which introduced six core principles: protecting autonomy, promoting human well-being and safety, ensuring transparency, fostering accountability, ensuring inclusiveness and equity, and promoting sustainable AI.[42,100] To operationalize these principles, the WHO has increasingly focused on readiness and the governance of frontier AI systems through its Global Strategy on Digital Health 2020–2027 and the 2024 AI Readiness Assessment for Public Health.[101]

Moreover, the WHO, in collaboration with ITU and WIPO, produces concrete outputs for technology-importing regions through the Global Initiative on AI for Health (GI-AI4H), supplying standardized benchmarking datasets, model regulatory texts for low- and middle-income countries, and post-market surveillance protocols for detecting data drift.[102,103] Furthermore, the WHO's 2025 Guidance on Large Multi-Modal Models issued over 40 specific recommendations, including prohibitions on autonomous clinical decision-making, mandatory watermarking of AI-generated outputs, third-party bias auditing, and the requirement of quarterly post-deployment performance reports during the first two years of use.[42] Recognizing the urgency of these challenges, a coalition of 15 global health organizations published an open letter in February 2026, urging the 194 Member States at the 79th World Health Assembly to adopt a binding resolution on global health data governance that would guarantee minimum de-identification standards, patient opt-out rights for commercial AI data use, and international data-sharing agreements.[104]

Faith-based Institutional Ethics, Global Health Networks, and Beyond

Beyond secular intergovernmental bodies, large-scale non-state and faith-based healthcare networks exercise significant operational influence over global health delivery. For example, as the largest non-government provider of health care services, Catholic church operates approximately 117,000 health facilities worldwide, representing a massive secondary tier of health infrastructure, particularly in low- and middle-income regions.[105,106] Thus, institutional ethics guidelines, such as the Leo XIV's 2026 papal encyclical Magnifica Humanitas, function not merely as moral doctrine, but as binding institutional and clinical governance standards in many scenarios.

By directly challenging the commercial concentration of compute power and opposing the monetization of patient data, these institutional frameworks impose operational boundaries on clinical deployment. They mandate that diagnostic and decision-support software deployed within their networks adhere to strict data-sovereignty rules, protect patient data from unconsented commercial monetization, and maintain direct human oversight in clinical workflows. In regions where state regulatory agencies lack enforcement capacity, such non-governmental ethical frameworks serve as a critical defense against the unvetted deployment of black-box algorithms, forcing technology providers to meet institutional equity and privacy standards as a prerequisite for clinical integration.

Crucially, the influence of such normative frameworks extends upstream, reaching beyond the point of care to directly impact the AI development pipeline. For instance, Magnifica Humanitas explicitly frames algorithmic design as a profoundly moral act, positing that "technological innovation can represent human participation in the divine act of creation." By elevating the act of development to a moral responsibility, accountability could be shifted onto the technology firms and developers themselves. It establishes a mandate that creators must bear a specific ethical and spiritual responsibility to embed transparency, accountability, and the common good from the outset of the design phase. Given the large population of these faith-based communities, this normative pressure exerts influence far beyond the walls of faith-based health networks,

establishing a broader ethical paradigm that challenges the global tech industry to design for human dignity before an algorithm ever reaches the clinic. For example, Anthropic's co-founder Chris Olah recently delivered a remark highlighting the faith-based ethics on the company's constitutional AI approach.[107] Therefore, by articulating these stringent ethical ceilings, non-governmental frameworks compensate for the operational fragmentation of state laws, providing a moral compass for clinicians, policymakers, and developers navigating an increasingly medical AI landscape.

Discussion and Policy Implications

The rapid development of medical AI and the deployment of highly scalable foundation models present an unprecedented opportunity to elevate global healthcare. However, as this policy analysis demonstrates, the global AI governance landscape remains dangerously fractured, not only in dominant technology-exporting superpowers, i.e., the US, the EU, and China, but also in technology-importing regions such as the Africa continent. The rest of this section discusses the related issues as well as policy implications (see Panel 2).

Reframing the Geopolitics of Digital Health in the Era of AI

A central contribution of this policy analysis is the shift from the conventional "Global North versus Global South" taxonomy that typically frames global health literature.[108,109] While such a taxonomy has been useful for analyzing traditional healthcare supply or infectious disease funding,[110,111] it fails to capture the structural realities of modern digital health infrastructure.[112,113] By categorizing the global landscape instead into dominant technology-exporting jurisdictions (the US, the EU, and China) and technology-importing regions (such as the African continent), this paper realigns the equity conversation around digital infrastructure control rather than geographic or legacy economic boundaries. This structural reframing demonstrates that the modern vector of health inequity in the AI era is not merely a lack of domestic financial resources, but an asymmetric dependency on foreign-hosted cloud compute, proprietary foundation models, and external regulatory architectures.

The Liability and Compliance Friction

Such a fragmentation introduces severe cross-border friction. The divergence in how jurisdictions handle data privacy, post-market adaptability, and liability allocation means that a medical AI tool optimized for clinical workflows in one region may face legal obsolescence or outright prohibition in another. As autonomous AI agents increasingly participate in multidisciplinary decision-making[33,35], the absence of a unified international liability framework threatens to leave both healthcare providers and consumers without clear recourse when algorithmic errors occur. Furthermore, fragmented data privacy architectures, e.g., the EU's strict special category protections vs. the US HIPAA exemptions, create barriers to building the diverse, multi-institutional datasets required to train generalizable algorithms.

Beyond structural liability, this regulatory fragmentation directly disrupts the socio-technical realities of medical AI development.[114,115] When jurisdictions enforce conflicting definitions of data privacy and mandatory human oversight, the friction cascades down into fundamental AI for healthcare engineering pipelines and user-centric design. For instance, AI developers are forced to engineer disjointed clinical data products by designing an interface that treats a clinician as a passive visual analyst in one market, while requiring an active, supervisory override architecture in another to satisfy local compliance. Consequently, this legal asymmetry neglects the critical socio-technical factors inherent in building complex healthcare algorithms, transforming what should be a scalable, user-centered clinical tool into a fragile, jurisdiction-specific engineering patchwork.

***Panel 2*: Policy Recommendations for Global Healthcare AI Governance**

1. For Technology-Exporting Regimes (US, EU, China)
    - **Extraterritorial Accountability**: Regulatory authorities must prohibit the export of medical algorithms that lack demographic validation in their target deployment populations, formally closing the loop on regulatory arbitrage.
    - **Regulating Infrastructure**: Policymakers must enforce public-interest oversight on global monopolies, encouraging fair licensing and non-extractive technology transfer.
2. For Technology-Importing Regions (African Continent/Global South)
    - **Statutory Institutional Mandate**: National governments must explicitly designate a primary statutory authority with the explicit duty to draft, enact, and continuously update health AI policy. Assigning clear institutional ownership eliminates inter-ministerial ambiguity, overcomes "draft inertia," and establishes direct state accountability for regulatory enforcement.
    - **Continental Approval Pathways**: Regional bodies like the African Medicines Agency (AMA) must operationalize a unified review mechanism and mutual recognition agreements to lower administrative barriers and block regulatory arbitrage.
    - **Sovereign AI Legislation**: Governments must enact data sovereignty laws to prevent unconsented commercial data extraction and incentivize localized compute infrastructure.
3. For Intergovernmental and Non-State Actors (WHO & Faith-Based Networks)
    - **Binding Global Standards**: Member States should urgently adopt WHO resolutions on health data governance at the World Health Assembly, ensuring minimum de-identification standards and patient opt-out rights.
    - **Operation as Enforcement**: Massive non-state health networks, such as the 117,000 global Catholic health facilities, must translate ethical frameworks like Magnifica Humanitas into operational criteria. In thin regulatory environments, these networks act as de facto enforcers by refusing to deploy AI that monetizes patient data or lacks human oversight.
4. Mechanisms for Multilateral Collaboration
    - **Cross-Border Sandboxes**: Jurisdictions should establish joint regulatory sandboxes, allowing exporters to validate algorithms on localized datasets in secure, sovereign environments.
    - **Regulatory Twinning**: Dominant agencies (FDA, NMPA, EMA) should formally partner with nascent continental bodies (AMA) to share review methodologies and fund regulatory capacity building.
    - **Co-Design Agreements**: Importing health ministries should make market access contingent upon exporters transferring foundational model capabilities and training local talent to maintain them, shifting from digital extraction to co-development.

Synthesizing the Governance Gap in Global Health

Ultimately, the operational fragmentation mapped in this analysis extends beyond legal friction; it represents a structural determinant of global health equity. Current global health literature increasingly emphasizes the necessity of decolonizing health systems and shifting structural power toward the Global South.[116,117] Within this context, the unvetted export of AI algorithms from dominant regulatory superpowers to technology-importing regions indeed functions as a form of digital colonialism,[118] particularly across the African continent. When jurisdictions with mature regulatory frameworks allow algorithms to be exported without demographic validation in target populations, they actively undermine the core tenets of health systems strengthening in low-resource settings. Furthermore, over-relying on foreign-hosted cloud infrastructure and proprietary foundation models threatens local health security. Consequently, the emergent push for Sovereign AI across the African continent should not be viewed merely as an economic or infrastructural ambition, but as a critical public health intervention. It is the digital equivalent of localized capacity building, ensuring that intelligent clinical tools are designed, validated, and governed by the populations they are intended to serve.

Limitations

We acknowledge several limitations in this policy analysis. First, as a macro-level evaluation of state and intergovernmental governance architectures, this paper does not present empirical, field-based clinical outcome data regarding the real-world deployment of specific algorithms. The socio-technical realities of how healthcare professionals interact with these fragmented compliance standards in daily clinical workflows require further ethnographic and qualitative investigation. Second, the regulatory landscape for healthcare AI is highly volatile. Several national strategies, particularly across the African continent, are currently in draft form or phased implementation, meaning the enforceability of these frameworks may shift rapidly. Finally, while we map the normative influence of major non-governmental actors, quantifying the exact impact of these ethical frameworks on commercial software procurement remains challenging and warrants dedicated empirical study.

Conclusion

The regulation of AI in healthcare is one of the most complex and dynamic areas of contemporary law and policy. The United States, the European Union, and China have each developed distinctive approaches that reflect their regulatory traditions, political contexts, and industrial priorities. The FDA's flexible, guidance-based approach emphasizes innovation and adaptability, using mechanisms like the PCCP to accommodate AI's iterative nature while maintaining safety oversight. The EU's AI Act represents the most comprehensive and systematic framework, establishing binding requirements across all high-risk AI applications; its precautionary, rights-based approach sets a global benchmark but imposes significant compliance burdens. Meanwhile, China's structured, rule-based approach combines detailed technical guidance with active international harmonization, reflecting dual ambitions to accelerate its domestic AI industry and influence global standards.

Africa presents a different picture. While 20 countries have developed or began to draft national AI strategies, many designating health as a priority, none have implemented binding risk classification, lifecycle change management, or post-market surveillance for adaptive AI systems. In the absence of an operative continental approval pathway, the gap between articulated AI ethical principles and enforceable regulatory capacity creates a severe risk of regulatory arbitrage. Addressing this vulnerability requires more than stronger continental coordination; it necessitates supporting the emergent drive toward Sovereign AI, ensuring that technology-importing regions secure the data sovereignty and localized infrastructure required to govern

algorithms safely.

Despite jurisdictional divergences, a principled convergence exists around risk-based classification, lifecycle oversight, data quality, transparency, and human oversight. Crucially, this convergence is not the product of governmental frameworks alone. Non-governmental actors, ranging from the WHO's technical instruments and the declarations of international medical associations to the Catholic Church's encyclical Magnifica Humanitas, articulate these same principles from humanitarian foundations. These entities reinforce regulatory norms and extend them into global institutional settings that formal state regulations often cannot penetrate.

The challenge ahead lies not merely in establishing these principles, but in resolving the operational and socio-technical friction of their implementation across borders. For developers, clinicians, and civil society, navigating this landscape requires bridging the gap between formal state governance and the essential moral infrastructure provided by non-governmental actors. Ultimately, ensuring the legitimacy of AI in healthcare depends on transitioning from fragmented national policies to a unified, human-centered global regulatory science.

Data Availability

The policy document archive and associated provenance metadata supporting this analysis are publicly available on Figshare at `https://doi.org/10.6084/m9.figshare.33400486`.

Contributors

KM, ZJ, and YW conceived the policy analysis, and drafted the primary manuscript. KM, PS, EHZ, WK, ZJ, and SM conducted the core regulatory text collecting, tracking, and analyses. All authors reviewed, edited, and approved the final manuscript for submission.

Declaration of Interests

We declare no competing interests.

Role of the Funding Source

This work is partially supported by National Natural Science Foundation under the grant 62076232. The corresponding authors had full access to all data elements and assumed final responsibility for the decision to submit for publication.

AI Declaration Statement

During the preparation of this manuscript, the authors utilized Claude Opus 4.8, Gemini 3.1 Pro, and GPT 5.6 Sol to support language polishing of the programmatic text blocks. Following the use of these tools, the authors thoroughly reviewed, critically evaluated, and edited the text to ensure absolute factual grounding and technical accuracy. The authors maintain full accountability and ownership over the final text, context, and structural analysis presented herein.

References

[1] The Lancet Digital Health. The rise of the agentic AI colleague. *The Lancet Digital Health* 2026; **8**(3): e101010.

[2] Truhn D, Azizi, S, Zou, J, et al. (2026). Artificial intelligence agents in cancer research and oncology. *Nature Reviews Cancer* 2026; **26**(4): 256-269.

[3] Angus DC, Khera R, Lieu T, et al. AI, Health, and Health Care Today and Tomorrow: The JAMA Summit Report on Artificial Intelligence. *JAMA* 2025; **334**(18):1650–1664.

[4] Lekadir K, Frangi AF, Porras AR, et al. FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare. *BMJ* 2025; 388.

[5] Kwet M. Digital colonialism: US empire and the new imperialism in the Global South. *Race & Class* 2019; **60**(4):3-26.

[6] Crum B. Brussels effect or experimentalism? The EU AI Act and global standard-setting. *Internet Policy Review* 2025; **14**(3):1-21.

[7] Ishkhanyan A. Governing AI across borders: corporate power, state sovereignty and global regulation. *Digital Policy, Regulation and Governance* 2026; **28**(3):280-95.

[8] Schneider Rasador G, Moreira Cunha A. The new security grey zone: Export controls, emerging technologies and US-China technological rivalry. *The Pacific Review* 2025; **38**(6):1020-48.

[9] European Union. General Data Protection Regulation (GDPR) 2018; Available from: https://gdpr-info.eu/.

[10] Cohen IG, Mello MM. HIPAA and protecting health information in the 21st century. *JAMA* 2018; **320**(3):6-7.

[11] Hassan Y. Governing algorithms from the South: A case study of AI development in Africa. *Ai & Society* 2023; **38**(4):1429-42.

[12] Ayana G, Dese K, Nemomssa HD, et al. Decolonizing global AI governance: assessment of the state of decolonized AI governance in Sub-Saharan Africa. *Royal Society Open Science* 2024; **11**(8):231994.

[13] Wakunuma K, Eke D. Africa, ChatGPT, and generative AI systems: Ethical benefits, concerns, and the need for governance. *Philosophies* 2024; **9**(3):80.

[14] Townsend BA. Governance-by-design as an enabler of AI in digital health in sub-Saharan Africa. *Law, Technology and Humans* 2026; **8**(1):8-22.

[15] Ogoh G, Eke D, Stahl BC. AI governance in North Africa: an ecosystems approach. *Science and Public Policy* 2026; **53**(2):258-76.

[16] Shonubi OA. The Open-Source Paradox: Africa's Digital Sovereignty and the Structural Limits of Artificial Intelligence Autonomy. *AI & Innovation* 2026; **1**(1):e70004.

[17] Barrett T, Okolo CT, Biira B, et al. African data ethics: a discursive framework for black decolonial AI. In*Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)* 2025 (pp. 334-349).

[18] Ministry of Information, Communications and the Digital Economy of the Republic of Kenya. Kenya Artificial Intelligence Strategy (2025–2030) 2025; Available from: https://ict.go.ke/.

[19] World Health Organization. Ethics and Governance of Artificial Intelligence for Health: Guidance on Large Multi-modal Models 2025; Available from: https://www.who.int/publications/i/item/9789240084759.

[20] Pope Leo XIV. Magnifica Humanitas: Encyclical Letter on Safeguarding the Human Person in the Time of Artificial Intelligence 2026; Available from: http://www.vatican.va/content/leo-xiv/en/encyclicals/documents/20260515-magnifica-humanitas.html.

[21] U.S. Food and Drug Administration. Artificial intelligence and machine learning (AI/ML)-enabled medical devices: 2025 device database statistics, FDA; 2025.

[22] National Medical Products Administration (NMPA). Statistics on approved AI medical device software products. Beijing, China: NMPA; 2025

[23] The Lancet Digital Health. Transforming digital pathology with AI. *The Lancet Digital Health* 2026; **8**(4): e101026.

[24] Moor M, Banerjee O, Abad ZS, et al. Foundation models for generalist medical artificial intelligence. *Nature* 2023; **616**(7956):259-65.

[25] Sheikh A, Anderson M, Albala S, et al. Health information technology and digital innovation for national learning health and care systems. *The Lancet Digital Health* 2021; **3**(6): e383–96

[26] Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nature Medicine* 2019; **25**(1): 44–56.

[27] McDuff D, Schaekermann M, Tu T, et al. Towards accurate differential diagnosis with large language models. *Nature* 2025; **642**: 451–457.

[28] Kanjee Z, Crowe B, Rodman A. Accuracy of a Generative Artificial Intelligence Model in a Complex Diagnostic Challenge. *JAMA* 2023; 330(1):78-80.

[29] Obermeyer Z, Powers B, Vogeli C, et al. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* 2019; **366**(6464): 447-453.

[30] Angus DC, Khera R, Lieu T, et al. AI, health, and health care today and tomorrow: the JAMA summit report on artificial intelligence. *JAMA*. 2025; **334**(18):1650-64.

[31] Esmaeilzadeh P. Challenges and strategies for wide-scale artificial intelligence (AI) deployment in healthcare practices: A perspective for healthcare organizations. *Artificial Intelligence in Medicine* 2024; **151**:102861.

[32] Widner K, Virmani S, Krause J, et al. Lessons learned from translating AI from development to deployment in healthcare. *Nature Medicine* 2023; **29**(6):1304-6.

[33] Cross JL, Choma MA, Onofrey JA. Bias in medical AI: implications for clinical decision-making. *PLOS Digital Health* 2024; **3**(11):e0000651.

[34] Pham T. Ethical and legal considerations in healthcare AI: innovation and policy for safe and fair use. *Royal Society Open Science* 2025; **12**(5):241873.

[35] Rajpurkar P, Chen E, Banerjee O, et al. AI in health and medicine. *Nature Medicine*. 2022; **28**(1):31-8.

[36] Schwalbe N, Wahl B. Artificial intelligence and the future of global health. *The Lancet* 2020; **395**(10236):1579-86.

[37] Pratt B. Research for Health Justice: an ethical framework linking global health research to health equity. *BMJ Global Health* 2021; **6**(2):e002921.

[38] Dychiao RG, Nazer L, Mlombwa D, et al. Artificial intelligence and global health equity. *BMJ* 2024; 387.

[39] Celi LA, Cellini J, Charpignon ML, Dee EC, Dernoncourt F, Eber R, Mitchell WG, Moukheiber L, Schirmer J, Situ J, Paguio J. Sources of bias in artificial intelligence that perpetuate healthcare disparities—A global review. *PLOS Digital Health* 2022; **1**(3):e0000022.

[40] Cross JL, Choma MA, Onofrey JA. Bias in medical AI: implications for clinical decision-making. *PLOS Digital Health* 2024; **3**(11):e0000651.

[41] Mittermaier M, Raza MM, Kvedar JC. Bias in AI-based models for medical applications: challenges and mitigation strategies. *NPJ Digital Medicine* 2023; **6**(1):113.

[42] Alderman JE, Palmer J, Laws E, et al. Tackling algorithmic bias and promoting transparency in health datasets: the STANDING Together consensus recommendations. *The Lancet Digital Health* 2025; **7**(1):e64-88.

[43] Han GR, Goncharov A, Eryilmaz M, Ye S, Palanisamy B, Ghosh R, Lisi F, Rogers E, Guzman D, Yigci D, Tasoglu S. Machine learning in point-of-care testing: innovations, challenges, and opportunities. *Nature Communications* 2025; **16**(1):3165.

[44] U.S. Food and Drug Administration. Marketing submission recommendations for a predetermined change control plan for artificial intelligence-enabled device software functions: Final guidance 2024.

[45] U.S. Food and Drug Administration. Premarket Approval (PMA) 2025.

[46] U.S. Food and Drug Administration. Total Product Life Cycle for Medical Devices (TPLC) 2023; Available from: `https://www.fda.gov/about-fda/cdrh-transparency/total-product-life-cycle-medical-devices`.

[47] U.S. Food and Drug Administration, Health Canada, and Medicines and Healthcare Products Regulatory Agency (MHRA). Good machine learning practice for medical device development: Guiding principles (GMLP) 2021; Available from: `https://www.gov.uk/government/publications/good-machine-learning-practice-for-medical-device-development-guiding-principles/`.

[48] European Parliament and Council of the European Union, Regulation (EU) 2024/1689, Artificial Intelligence Act 2024; Available from: `https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng`.

[49] Laux J, Wachter S, Mittelstadt B. Trustworthy artificial intelligence and the European Union AI act: On the conflation of trustworthiness and acceptability of risk. *Regulation & Governance* 2024; **18**(1):3-32.

[50] Busch F, Kather JN, Johner C, et al. Navigating the European union artificial intelligence act for healthcare. *NPJ Digital Medicine*. 2024; **7**(1):210.

[51] European Parliament and Council of the European Union, Regulation (EU) 2017/745, Regulation on Medical Devices 2017; Available from: `https://eur-lex.europa.eu/eli/reg/2017/745/oj/eng`.

[52] European Parliament and Council of the European Union, Regulation (EU) 2017/746, Regulation on in vitro Diagnostic Medical Devices 2017; Available from: https://eur-lex.europa.eu/eli/reg/2017/746/oj/eng.

[53] Joint Artificial Intelligence Board and Medical Device Coordination Group, AIB 2025-1/MDCG 2025-6, Interplay between the Medical Devices Regulation (MDR) & In vitro Diagnostic Medical Devices Regulation (IVDR) and the Artificial Intelligence Act (AIA); Available from: https://health.ec.europa.eu/document/download/b78a17d7-e3cd-4943-851d-e02a2f22bbb4_en.

[54] Ziosi M, Mökander J, Novelli C et al. The EU AI liability directive (AILD): bridging information gaps. *European Journal of Law and Technology* 2023; **14**(3).

[55] Han Y, Ceross A, Bergmann J. Regulatory frameworks for AI-enabled medical device software in China: comparative analysis and review of implications for global manufacturer. *JMIR AI* 2024; **3**:e46871.

[56] Zhang L, Yan J. Approval of AI-based medical devices in China from 2020 to 2025: Retrospective analysis. *JMIR Medical Informatics* 2026; **14**(1):e85538.

[57] Liu Y, Yu W, Dillon T. Regulatory responses and approval status of artificial intelligence medical devices with a focus on China. *NPJ Digital Medicine* 2024; **7**(1):255.

[58] Artificial Intelligence Medical Device Standardization Technical Committee (SMD/TU 002). AI medical device standardization work report. National Medical Products Administration (NMPA), Beijing, China, 2025.

[59] Noesselt N. Fragmented Global AI Governance? Chinese AI Leadership Dreams and Africa's Quest for Digital Autonomy. Global Policy 2026; Online first: https://doi.org/10.1111/1758-5899.70199.

[60] Roberts H, Cowls J, Morley J, Taddeo M, Wang V, Floridi L. The Chinese approach to artificial intelligence: an analysis of policy, ethics, and regulation. *AI & society* 2021; **36**(1):59-77.

[61] Asiedu MN, Dieng A, Haykel I, Rostamzadeh N, Pfohl S, Nagpal C, Nagawa M, Oppong A, Koyejo S, Heller K. The case for globalizing fairness: a mixed methods study on colonialism, AI, and health in Africa. In*Proceedings of the 4th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)* 2024 (pp. 1-24).

[62] Victor A. Artificial intelligence in global health: An unfair future for health in Sub-Saharan Africa?. *Health Affairs Scholar* 2025; **3**(2):qxaf023.

[63] Tubaro P, Casilli AA, Cornet M, et al. Where does AI come from? A global case study across Europe, Africa, and Latin America. *New Political Economy* 2025; **30**(3):359-72.

[64] Diallo K, Smith J, Okolo CT, Nyamwaya D, Kgomo J, Ngamita R. Case studies of AI policy development in Africa. *Data & Policy*. 2025; **7**:e15.

[65] Holden K, Harsh M. On pipelines, readiness and annotative labour: Political geographies of AI and data infrastructures in Africa. *Political Geography* 2024; **113**:103150.

[66] Mateen BA, Ordish J, Reid MJ. Your AI tool is subject to medical device regulation. *Nature Health* 2026; **1**(1):9-10.

[67] Ministry of Digitalisation and Innovation of the Republic of Benin. Stratégie Nationale d'Intelligence Artificielle et des Mégadonnées (SNIAM) 2023–2027 (in French). 2023; Available from: https://innovation.gouv.bj/publications/documents/strategie-nationale-d-intelligence-artificielle-et-des-megadonnees-2023-2027.

[68] Ministry of Communications and Digitalisation of the Republic of Ghana. National Artificial Intelligence Strategy 2023-2033 (updated as National AI Strategy 2025-2035 in Dec. 2025). 2023; Available from: https://aighana.net/ghana-national-ai-strategy/.

[69] Ministry of ICT and Innovation of the Republic of Rwanda, The National AI Policy. 2023; Available from: https://www.minict.gov.rw/ai-policy.

[70] Council of Ministers of the Federal Democratic Republic of Ethiopia, National Artificial Intelligence Policy. 2024; Available from: http://www.mint.gov.et/policy-and-strategy/-/document_library/CNlICzpide2L/view_file/240851.

[71] Federal Ministry of Communications, Innovation and Digital Economy of the Federal Republic of Nigeria. National Artificial Intelligence Strategy (NAIS). 2025; Available from: https://fmcide.gov.ng/initiative/nais/.

[72] The National Council of Artificial Intelligence, The Arab Republic of Egypt. Egypt National Artificial Intelligence Strategy–Second Edition (2025-2030). 2025; Available from: https://ai.gov.eg/SynchedFiles/en/Resources/AIstrategyEnglish16-1-2025-1.pdf

[73] The Egyptian Center for Responsible Artificial Intelligence (ECRAI). Guide to Egypt's National AI Governance Framework. 2026; Available from: https://ai.gov.eg/SynchedFiles/ar/Resources/.

[74] Ministry of ICT, Postal and Courier Services (MICTPCS) of the Republic of Zimbabwe. Zimbabwe National Artificial Intelligence Strategy. 2026; Available from: https://zimaistrategy.net/.

[75] Working Group on AI of the Government of Mauritius. Mauritius Artificial Intelligence Strategy. 2018; Available from: https://treasury.govmu.org/Documents/Strategies/MauritiusAIStrategy.pdf.

[76] Agence nationale de promotion de la recherche scientifique (ANPR), Tunisia. National AI Strategy: Unlocking Tunisia's capabilities potential. 2019; Available from: https://www.aipolicy.africa/national-strategies/bb3c8413-e77a-4427-aa5d-568942454c6b.

[77] Ministry of Communication, Telecommunications and Digital Economy of the Republic of Senegal. Senegal National Artificial Intelligence Strategy and Road Map (in French). 2023; Available from: https://www.archives.sn/docs/strategies/srategie-nationale-intelligence-artificielle.

[78] Ministry of Digital Transition and Administration Reform of the Kingdom of Morocco. Digital Morocco 2030. 2024; Available from: https://maroc.ma/en/strategies-and-public-policies/digital-morocco-2030.

[79] Ministry of Technology and Science of the Republic of Zambia. National Artificial Intelligence Strategy (2024-2026) 2024; Available from: https://www.mots.gov.zm/wp-content/uploads/2025/02/Zambia-Ai-Strategy-Book-option-2.pdf.

[80] Ministry of Posts and Telecommunications. National Artificial Intelligence Strategy (SNIA), Press Release. 2025; Available from:https://www.cameroononline.org/cameroon-unveils-seven-pillar-ai-strategy-to-become-africas-leading-hub-by-2040/.

[81] Ministry of Digital Transition and Technological Innovation of the Republic of Côte d'Ivoire. Stratégie Nationale de l'Intelligence Artificielle à l'horizon 2030 (in French). 2025; Available from: https://www.telecom.gouv.ci/new/uploads/publications/174196670372.pdf.

[82] Department of Communications and Digital Technologies of the Republic of South Africa, South Africa National AI Policy Framework. 2024; Available from: https://www.dcdt.gov.za/sa-national-ai-policy-framework.html

[83] Department of Communications and Digital Technologies of the Republic of South Africa, Draft South Africa National Artificial Intelligence Policy. 2026; Available from: https://www.gov.za/sites/default/files/gcis_document/202604/54477gen3880.pdf

[84] Ministry of Information, Communication, and Technology of the United Republic of Tanzania. Guidelines for the Secure and Ethical Use of Artificial Intelligence in Tanzania (Draft Version One). 2025; Available from: https://www.mawasiliano.go.tz/uploads/documents/sw-1749982790-Guidelines%20for%20AI%20ethical%20USE%20Guideline%20MICIT%202025ver.pdf

[85] Ministry of ICT & National Guidance of the the Republic of Uganda. Uganda National AI Landscape Assessment. 2024; Available from: https://ict.go.ug/site/documents/uganda-national-ai-landscape-assessment.pdf.

[86] Ministry of Communications and Innovation of the Republic of Botswana. title: 2026/2027 Recurrent and Development Budget Proposal. 2026; Available from: https://www.botswanaspeaks.gov.bw/media/COMMITTEE%20OF%20SUPPPLY%20SPEECHES/2026%20MCI%20COMMITTEE%20OF%20SUPPLY%20-%20%2016%2003%202026%20REVISED%20FINAL%20%28%20Hon%20Minister%29%201.pdf.

[87] National Commission on Research, Science and Technology of Namibia. Namibia AI Ecosystem. 2026; Available from: https://aiecosystem.ncrst.na/.

[88] African Union. Continental Artificial Intelligence Strategy. 2024; Available from: https://au.int/en/documents/20240809/continental-artificial-intelligence-strategy.

[89] African Union. Treaty for the establishment of African Medicines Agency. 2021; Available from: https://au.int/sites/default/files/treaties/36892-treaty-0069_-_ama_treaty_e.pdf.

[90] Sajida S, Pratiwi RD. Laying the Groundwork for Artificial Intelligence Policy: A Preliminary Assessment of Indonesian Governance Using UNESCO's RAM Framework. *Public Service and Governance Journal* 2025; **6**(1):309-27.

[91] Mbewe A, Sarkwah HN, Ijaodola O, et al. The role of digital health and artificial intelligence in improving the reach and effectiveness of HIV prevention in Africa. *The Lancet Global Health* 2026; **14**(1):e131-42.

[92] Cassava Technologies. Cassava scales African AI Infrastructure with NVIDIA-Powered AI Factories to accelerate sovereign data capabilities. 2026; Available from: https://www.cassava.ai/.

[93] Shead C. Intron Health gets backing for its speech-recognition tool that recognizes African accents. *TechCrunch* 2024; Available from: https://techcrunch.com/2024/07/25/intron-health-raises-1-6m-pre-seed/.

[94] Tsanni A. This company is building AI for African languages. *MIT Technology Review* 2023; Available from: https://www.technologyreview.com/2023/11/17/1083637/lelapa-ai-african-languages-vulavula/.

[95] Kshetri N. Empowering African development: The transformative potential of artificial intelligence across sectors. *IT Professional* 2024; **26**(4), 97-103.

[96] World Medical Association. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human participants. *JAMA* 2025; **333**(1):71-4.

[97] Junior HL. AI ethics in medical research: the 2024 Declaration of Helsinki. *The Lancet* 2024; **404**(10467):2048-9.

[98] Asadollahi A. Governing by design: Algorithmic normativity, clinical standards, and health policy implications of AI in healthcare. *AI and Ethics* 2026; **6**(1):119.

[99] World Health Organization. Emergency use of unproven clinical interventions outside clinical trials: ethical considerations. World Health Organization; 2022; Available from: `https://www.who.int/publications/i/item/9789240041745`.

[100] Templin T, Perez MW, Sylvia S et al. Addressing 6 challenges in generative AI for digital health: a scoping review. *PLOS digital health* 2024; **3**(5):e0000503.

[101] World Health Organization. Global strategy on digital health 2020-2027. World Health Organization 2025; Available from: `https://www.who.int/publications/i/item/9789240116870`.

[102] World Health Organization, International Telecommunication Union, and World Intellectual Property Organization. Global Initiative on AI for Health (GI-AI4H). 2023; Available from: `https://www.who.int/initiatives/global-initiative-on-ai-for-health`.

[103] Muralidharan V, Ng MY, AlSalamah S, et al. Global Initiative on AI for Health (GI-AI4H): strategic priorities advancing governance across the United Nations. *NPJ Digital Medicine*. 2025; **8**(1):219.

[104] Mothci D, Duarte G, Adetifa I, et al. Open letter to WHO Member States calling for a global framework on health data governance ahead of the 79th World Health Assembly. 2026; Available from: `https://unu.edu/global-health/article/open-letter-ahead-158th-who-executive-board-and-79th-world-health-assembly-senior`.

[105] Agnew J. Deus vult: The geopolitics of the Catholic Church. *Geopolitics* 2010; **15**(1):39-61.

[106] Calderisi R. Earthly Mission: The Catholic Church and world development. Yale University Press 2013.

[107] Olah C. Remarks on Pope Leo XIV's encyclical "Magnifica Humanitas". 2026; Available from: `https://www.anthropic.com/news/chris-olah-pope-leo-encyclical`.

[108] Kim JU, Oleribe O, Njie R, Taylor-Robinson SD. A time for new north–south relationships in global health. International Journal of General Medicine. 2017; **10**:401-8.

[109] Pai M, Bandara S, Kyobutungi C. Shifting power in global health will require leadership by the Global South and allyship by the Global North. *The Lancet* 2024; **404**(10464):1711-3.

[110] Graham HR, King C, Rahman AE, Kitutu FE, et al. Reducing global inequities in medical oxygen access: the Lancet Global Health Commission on medical oxygen security. *The Lancet Global Health* 2025; **13**(3):e528-84.

[111] Mendelson M, Laxminarayan R, Limmathurotsakul D, et al. Antimicrobial resistance and the great divide: inequity in priorities and agendas between the Global North and the Global South threatens global mitigation of antimicrobial resistance. *The Lancet Global Health* 2024; **12**(3):e516-21.

[112] Burkhardt A. Infrastructural perspectives on artificial intelligence—on the implications of AI infrastructures for global (in) justice. InTransnational infrastructures and the transformation of global orders 2025 May 23 (pp. 81-96). Cham: Springer Nature Switzerland.

[113] Tan TF, Thirunavukarasu AJ, Jin L, et al. Artificial intelligence and digital health in global eye health: opportunities and challenges. *The Lancet Global Health* 2023; **11**(9):e1432-43.

[114] Dako F, Moraes FY, Doo F, et al. Digital health and artificial intelligence innovations for oncology in sub-Saharan Africa. *The Lancet Oncology* 2025; **26**(10):e547-57.

[115] Kickbusch I, Piselli D, Agrawal A, et al. The Lancet and Financial Times Commission on governing health futures 2030: growing up in a digital world. *The Lancet* 2021; **398**(10312):1727-76.

[116] Png MT. At the tensions of south and north: Critical roles of global south stakeholders in AI governance. In*Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT)* 2022 (pp. 1434-1445).

[117] Taylor L, Broeders D. In the name of development: Power, profit and the datafication of the global south. *Geoforum* 2015; **64**:229-37.

[118] Kwet M. Digital colonialism: US empire and the new imperialism in the Global South. *Race & Class* 2019; **60**(4):3-26.